

Assignment - MongoDB

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IT-2234(P) - Web Services and Server Technologies
Department of Physical Science
University of Vavuniya
Assignment – MongoDB

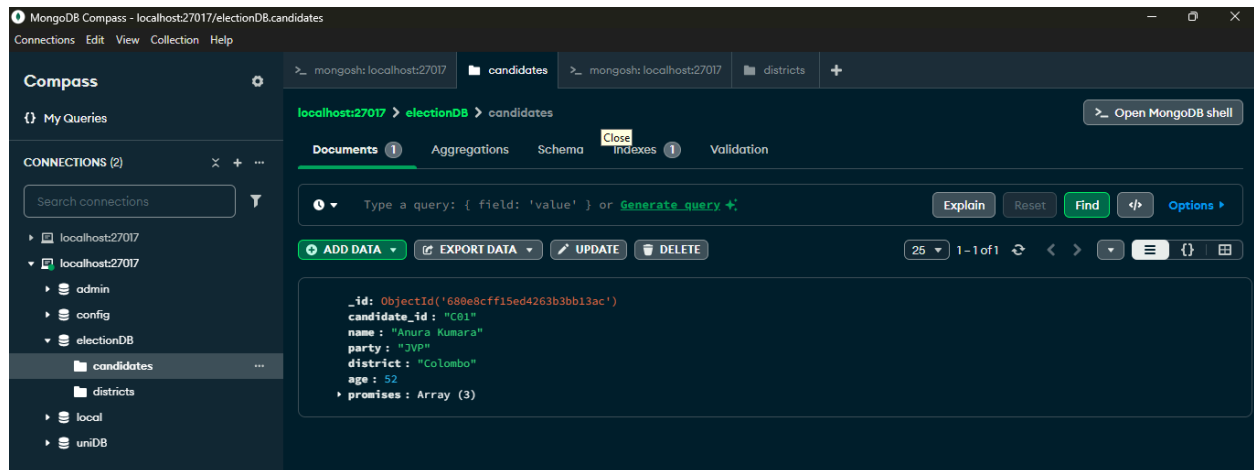
Question 01

1. Create a database, collection and insert data.

Code:

```
use electionDB

db.candidates.insertOne({
  "candidate_id": "C1001",
  "name": "Anura Kumara",
  "party": "JVP",
  "district": "Colombo",
  "age": 52,
  "promises": ["Free Education", "Healthcare", "Jobs"]
})
```



2. Insert many candidates at once.

Code:

```
db.candidates.insertMany([
```

```
{
  "candidate_id": "C02",
  "name": "Sajith Premadasa",
  "party": "SJB",
  "district": "Colombo",
  "age": 56,
  "promises": ["Housing", "Education"]
},

{
  "candidate_id": "C03",
  "name": "Gotabaya Rajapaksa",
  "party": "SLPP",
  "district": "Kandy",
  "age": 75,
  "promises": ["Security", "Infrastructure"]
},

{
  "candidate_id": "C04",
  "name": "Ruwan Wijewardene",
  "party": "UNP",
  "district": "Gampaha",
  "age": 48,
  "promises": ["Youth power", "Technology", "Education"]
}
```

])

Registration Number: 2021/ICT/59

```
>_ mongosh:localhost:27017  candidates >_ mongosh:localhost:27017  districts +
>_MONGOSH
> db.candidates.insertMany([
  {
    "candidate_id": "C02",
    "name": "Sajith Premadasa",
    "party": "SJB",
    "district": "Colombo",
    "age": 56,
    "promises": ["Housing", "Education"]
  },
  {
    "candidate_id": "C03",
    "name": "Gotabaya Rajapaksa",
    "party": "SLPP",
    "district": "Kandy",
    "age": 75,
    "promises": ["Security", "Infrastructure"]
  },
  {
    "candidate_id": "C04",
    "name": "Ruwan Wijewardene",
    "party": "UNP",
    "district": "Gampaha",
    "age": 48,
    "promises": ["Youth power", "Technology", "Education"]
  }
])
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('680e906ae8cc37f3ccf0227a'),
    '1': ObjectId('680e906ae8cc37f3ccf0227b'),
    '2': ObjectId('680e906ae8cc37f3ccf0227c')
  }
}
```

localhost:27017 > electionDB > candidates

Documents 1 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#)

[ADD DATA](#) [EXPORT DATA](#) [UPDATE](#) [DELETE](#)

Document 1

```
{
  "_id": ObjectId('680e8cfff15ed4263b3bb13ac'),
  "candidate_id": "C01",
  "name": "Anura Kumara",
  "party": "JVP",
  "district": "Colombo",
  "age": 52,
  "promises": Array (3)
}
```

Document 2

```
{
  "_id": ObjectId('680e906ae8cc37f3ccf0227a'),
  "candidate_id": "C02",
  "name": "Sajith Premadasa",
  "party": "SJB",
  "district": "Colombo",
  "age": 56,
  "promises": Array (2)
}
```

Document 3

```
{
  "_id": ObjectId('680e906ae8cc37f3ccf0227b'),
  "candidate_id": "C03",
  "name": "Gotabaya Rajapaksa",
  "party": "SLPP",
  "district": "Kandy",
  "age": 75,
  "promises": Array (2)
}
```

Document 4

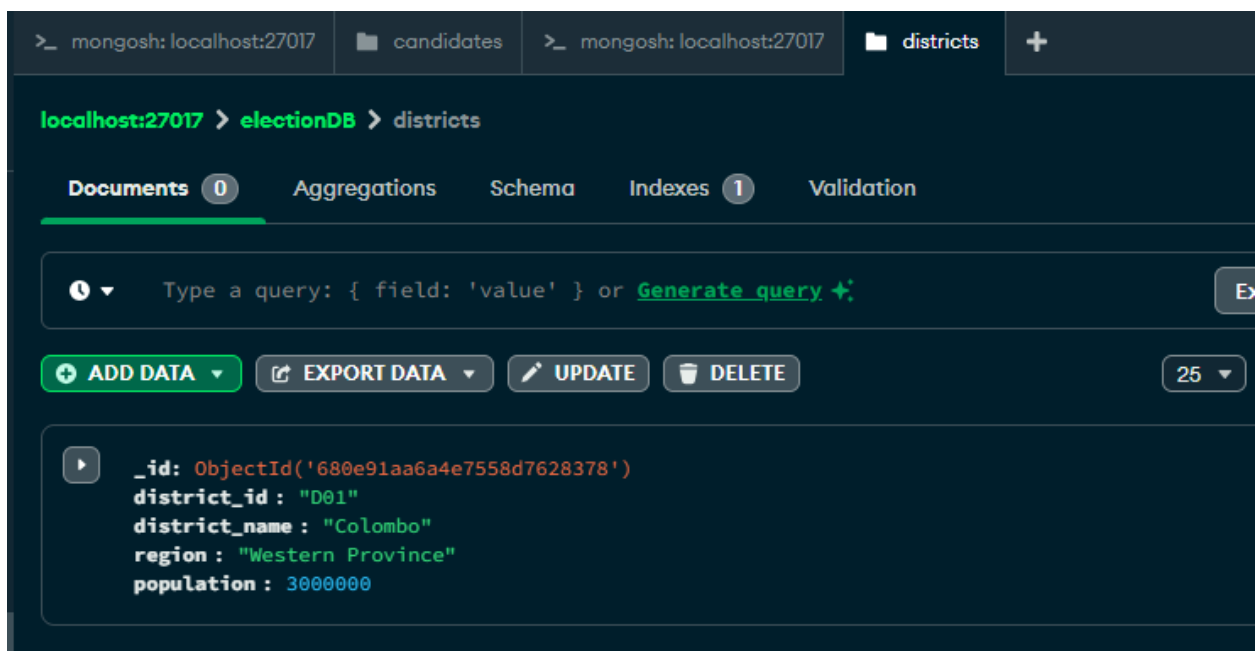
```
{
  "_id": ObjectId('680e906ae8cc37f3ccf0227c'),
  "candidate_id": "C04",
  "name": "Ruwan Wijewardene",
  "party": "UNP",
  "district": "Gampaha",
  "age": 48,
  "promises": Array (3)
}
```

3. Create a districts collection and insert data.

Code:

```
db.districts.insertOne({
  "district_id": "D01",
  "district_name": "Colombo",
  "region": "Western Province",
  "population": 3000000
})
```

```
> db.districts.insertOne({
  "district_id": "D01",
  "district_name": "Colombo",
  "region": "Western Province",
  "population": 3000000
})
< {
  acknowledged: true,
  insertedId: ObjectId('680e91aa6a4e7558d7628378')
}
electionDB>
```



4. Insert many districts at once.

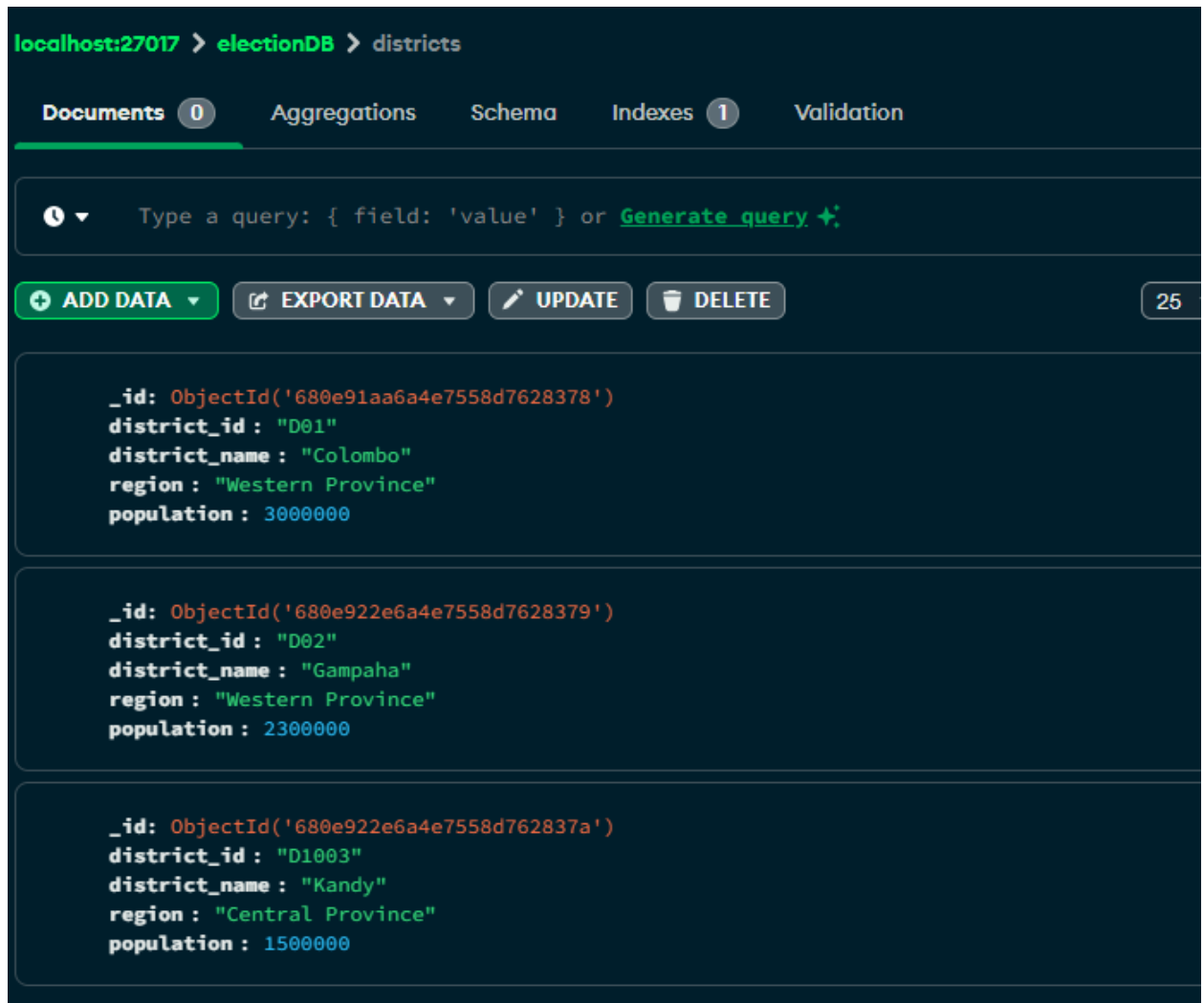
Query:

```
db.districts.insertMany([
  {
    "district_id": "D02",
    "district_name": "Gampaha",
    "region": "Western Province",
    "population": 2300000
  },

  {
    "district_id": "D1003",
    "district_name": "Kandy",
    "region": "Central Province",
    "population": 1500000
  }
])
```

```
db.districts.insertMany([
  {
    "district_id": "D02",
    "district_name": "Gampaha",
    "region": "Western Province",
    "population": 2300000
  },

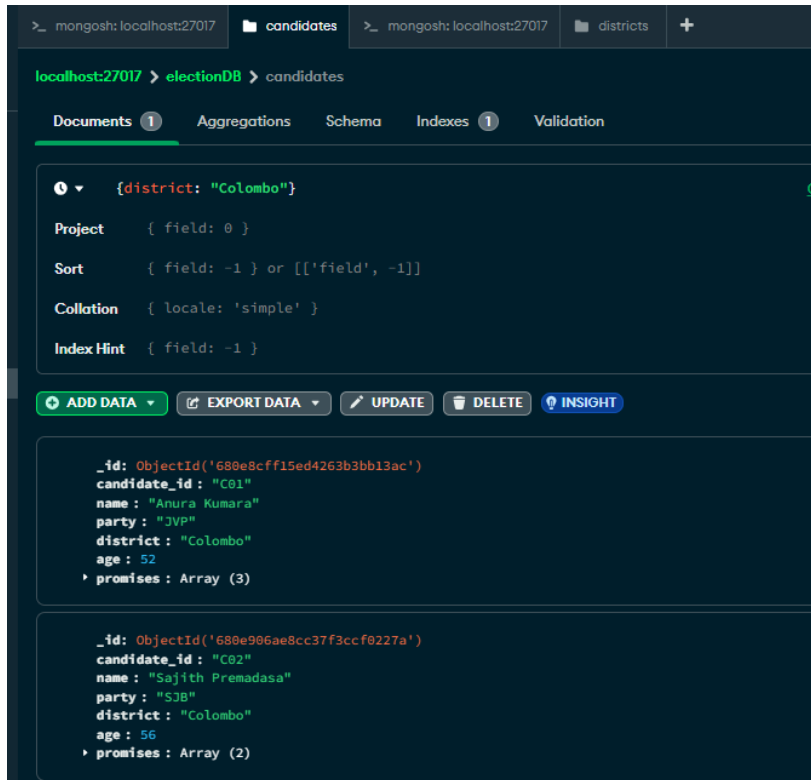
  {
    "district_id": "D1003",
    "district_name": "Kandy",
    "region": "Central Province",
    "population": 1500000
  }
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('680e922e6a4e7558d7628379'),
    '1': ObjectId('680e922e6a4e7558d762837a')
  }
}
lectionDB> |
```



5. List all candidates from the Colombo district

Query:

```
{district: "Colombo"}
```



Shell Query:

```
db.candidates.find({ "district": "Colombo" })
```

```
> db.candidates.find({ "district": "Colombo" })
< {
  _id: ObjectId('680e8cff15ed4263b3bb13ac'),
  candidate_id: 'C01',
  name: 'Anura Kumara',
  party: 'JVP',
  district: 'Colombo',
  age: 52,
  promises: [
    'Free Education',
    'Healthcare',
    'Jobs'
  ]
}
{
  _id: ObjectId('680e906ae8cc37f3ccf0227a'),
  candidate_id: 'C02',
  name: 'Sajith Premadasa',
  party: 'SJB',
  district: 'Colombo',
  age: 56,
  promises: [
    'Housing',
    'Education'
  ]
}
electionDB>
```


6. Find all districts in the Western Province.

Query:

```
{region: "Western Province"}
```

The screenshot shows the MongoDB Compass interface. At the top, there are tabs for different databases: 'candidates' and 'districts'. The 'districts' tab is selected. Below the tabs, the path 'localhost:27017 > electionDB > districts' is displayed. The 'Documents' tab is active, showing 0 documents. The query '{region: "Western Province"}' is entered in the search bar. Below the query, there are settings for Project, Sort, Collation, and Index Hint. The 'ADD DATA' button is highlighted. Below the buttons, two document snippets are shown, each representing a district in the Western Province.

```
{region: "Western Province"}
```

Project { field: 0 }

Sort { field: -1 } or [['field', -1]]

Collation { locale: 'simple' }

Index Hint { field: -1 }

ADD DATA **EXPORT DATA** **UPDATE** **DELETE** **INSIGHT**

```
_id: ObjectId('680e91aa6a4e7558d7628378')
district_id: "D01"
district_name: "Colombo"
region: "Western Province"
population: 3000000
```

```
_id: ObjectId('680e922e6a4e7558d7628379')
district_id: "D02"
district_name: "Gampaha"
region: "Western Province"
population: 2300000
```

Shell Query:

```
db.districts.find({ "region": "Western Province" })
```

```
> db.districts.find({ "region": "Western Province" })
< {
  _id: ObjectId('680e91aa6a4e7558d7628378'),
  district_id: 'D01',
  district_name: 'Colombo',
  region: 'Western Province',
  population: 3000000
}
{
  _id: ObjectId('680e922e6a4e7558d7628379'),
  district_id: 'D02',
  district_name: 'Gampaha',
  region: 'Western Province',
  population: 2300000
}
electionDB>
```

7. Show only the name and party of all candidates.

Query:

Project → {name:1, party:1, _id:0}

The screenshot shows the MongoDB Compass interface. At the top, there are tabs for 'candidates' and 'districts'. The 'candidates' tab is active. Below the tabs, the path 'localhost:27017 > electionDB > candidates' is displayed. The 'Documents' tab is selected. A query is entered in the 'Project' field: `{name:1, party:1, _id:0}`. Below the query field, there are options for 'Sort', 'Collation', and 'Index Hint'. An 'EXPORT DATA' button is visible. The results are displayed in a list of documents, each showing the 'name' and 'party' fields.

name	party
Anura Kumara	JVP
Sajith Premadasa	SJB
Gotabaya Rajapaksa	SLPP
Ruwan Wijewardene	UNP

Shell Query:

```
db.candidates.find({}, {name:1, party:1, _id:0})
```

```
> db.candidates.find({}, {name:1, party:1, _id:0})
< {
  name: 'Anura Kumara',
  party: 'JVP'
}
{
  name: 'Sajith Premadasa',
  party: 'SJB'
}
{
  name: 'Gotabaya Rajapaksa',
  party: 'SLPP'
}
{
  name: 'Ruwan Wijewardene',
  party: 'UNP'
}
electionDB>
```

8. Find the party of the candidate named "Anura Kumara".

Query:

```
db.candidates.find({ "name": "Anura Kumara" }, { "party": 1,
_id: 0 })
```

```
> db.candidates.find({ "name": "Anura Kumara" }, { "party": 1, _id: 0 })
< {
  party: 'JVP'
}
electionDB>
```

9. Find candidates who are older than 55 years.

Query:

```
{age: {$gt: 55}}
```

The screenshot shows the MongoDB Compass interface. At the top, there are tabs for 'candidates' and 'districts'. The main view is for the 'candidates' collection in the 'electionDB' database on 'localhost:27017'. The 'Documents' tab is selected, showing a query filter: `{age: {$gt: 55}}`. Below the filter, there are options for Project, Sort, Collation, and Index Hint. A toolbar contains buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', 'DELETE', and 'INSIGHT'. The query results are displayed in a list of documents. The first document is for Sajith Premadasa, and the second is for Gotabaya Rajapaksa.

Document
<pre>{ "_id": ObjectId("680e906ae8cc37f3ccf0227a"), "candidate_id": "C02", "name": "Sajith Premadasa", "party": "SJB", "district": "Colombo", "age": 56, "promises": Array (2) }</pre>
<pre>{ "_id": ObjectId("680e906ae8cc37f3ccf0227b"), "candidate_id": "C03", "name": "Gotabaya Rajapaksa", "party": "SLPP", "district": "Kandy", "age": 75, "promises": Array (2) }</pre>

Shell Query:

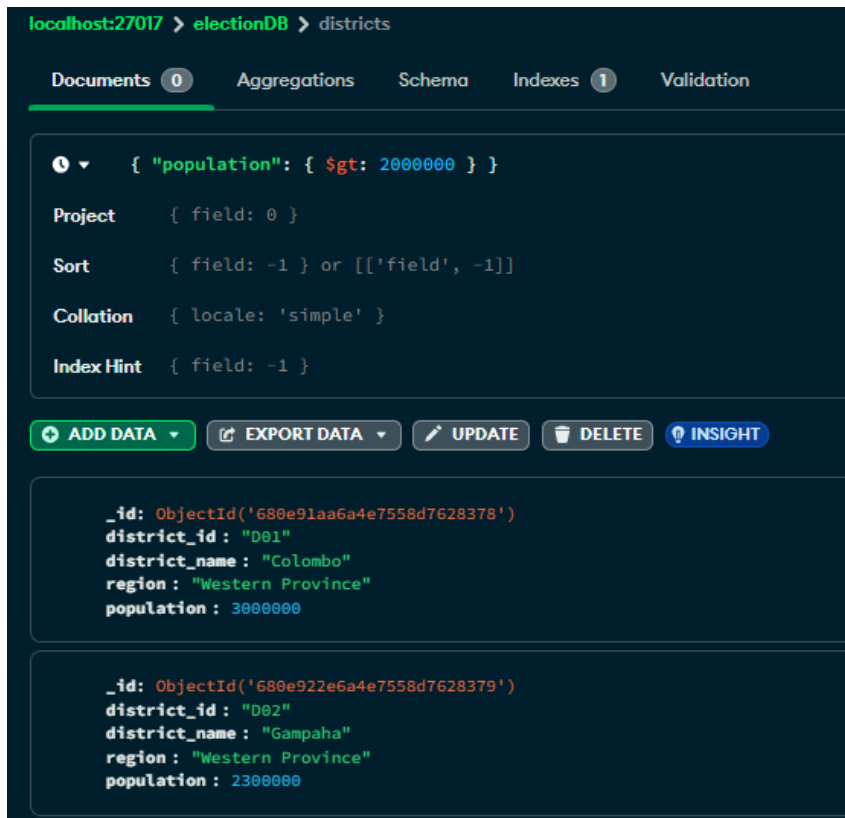
```
db.candidates.find({age: {$gt: 55}})
```

```
> db.candidates.find({age: {$gt: 55}})
< {
  _id: ObjectId('680e906ae8cc37f3ccf0227a'),
  candidate_id: 'C02',
  name: 'Sajith Premadasa',
  party: 'SJB',
  district: 'Colombo',
  age: 56,
  promises: [
    'Housing',
    'Education'
  ]
}
{
  _id: ObjectId('680e906ae8cc37f3ccf0227b'),
  candidate_id: 'C03',
  name: 'Gotabaya Rajapaksa',
  party: 'SLPP',
  district: 'Kandy',
  age: 75,
  promises: [
    'Security',
    'Infrastructure'
  ]
}
electionDB>
```

10. Find the districts whose population is greater than 2 million

Query:

```
{ "population": { $gt: 2000000 } }
```



Shell Query:

```
db.districts.find({ "population": { $gt: 2000000 } })
```



11. Find candidates who have "Education" mentioned in their promises.

Shell Query:

```
db.candidates.find({ "promises": { $in: ["Education"] } })
```

```
> db.candidates.find({ "promises": { $in: ["Education"] } })
< {
  _id: ObjectId('680e906ae8cc37f3ccf0227a'),
  candidate_id: 'C02',
  name: 'Sajith Premadasa',
  party: 'SJB',
  district: 'Colombo',
  age: 56,
  promises: [
    'Housing',
    'Education'
  ]
}
{
  _id: ObjectId('680e906ae8cc37f3ccf0227c'),
  candidate_id: 'C04',
  name: 'Ruwan Wijewardene',
  party: 'UNP',
  district: 'Gampaha',
  age: 48,
  promises: [
    'Youth power',
    'Technology',
    'Education'
  ]
}
electionDB> |
```

12. Find candidates from Colombo sorted by age (ascending).

Query:

Filter -> { "district": "Colombo" }

Sort -> { "age": 1 }

The screenshot shows the MongoDB Compass interface. At the top, there are tabs for 'candidates' and 'districts'. The main view is for the 'electionDB' database, 'candidates' collection. The 'Documents' tab is selected, showing a list of documents. The first document is expanded, showing its fields: '_id', 'candidate_id', 'name', 'party', 'district', 'age', and 'promises'. The second document is also expanded, showing its fields: '_id', 'candidate_id', 'name', 'party', 'district', 'age', and 'promises'. The interface includes a search bar at the top, a filter bar, and a toolbar with buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', 'DELETE', and 'INSIGHT'. The bottom of the interface shows the raw JSON data for the selected documents.

```
{ "district": "Colombo" }
```

Project: {}

Sort: { "age": 1 }

Collation: { locale: 'simple' }

Index Hint: { field: -1 }

ADD DATA **EXPORT DATA** **UPDATE** **DELETE** **INSIGHT**

```
{
  "_id": ObjectId('680e8cfff15ed4263b3bb13ac'),
  "candidate_id": "C01",
  "name": "Anura Kumara",
  "party": "JVP",
  "district": "Colombo",
  "age": 52,
  "promises": Array (3)
}
```

```
{
  "_id": ObjectId('680e906ae8cc37f3ccf0227a'),
  "candidate_id": "C02",
  "name": "Sajith Premadasa",
  "party": "SJB",
  "district": "Colombo",
  "age": 56,
  "promises": Array (2)
}
```

Shell Query:

```
db.candidates.find({ "district": "Colombo" }).sort({ "age": 1
})
```



```
> db.candidates.find({ "district": "Colombo" }).sort({ "age": 1 })
< {
  _id: ObjectId('680e8cff15ed4263b3bb13ac'),
  candidate_id: 'C01',
  name: 'Anura Kumara',
  party: 'JVP',
  district: 'Colombo',
  age: 52,
  promises: [
    'Free Education',
    'Healthcare',
    'Jobs'
  ]
}
{
  _id: ObjectId('680e906ae8cc37f3ccf0227a'),
  candidate_id: 'C02',
  name: 'Sajith Premadasa',
  party: 'SJB',
  district: 'Colombo',
  age: 56,
  promises: [
    'Housing',
    'Education'
  ]
}
electionDB>|
```

13. Sort the districts by population.

Query:

Sort -> { "population": 1 }

>_ mongosh: localhost:27017 candidates >_ mongosh: localhost:27017 districts +

localhost:27017 > electionDB > districts

Documents 0 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) ↗

Project { field: 0 }

Sort { "population": 1 }

Collation { locale: 'simple' }

Index Hint { field: -1 }

+ ADD DATA ▾ EXPORT DATA ▾ UPDATE DELETE INSIGHT

```
_id: ObjectId('680e922e6a4e7558d762837a')
district_id: "D1003"
district_name: "Kandy"
region: "Central Province"
population: 1500000
```

```
_id: ObjectId('680e922e6a4e7558d7628379')
district_id: "D02"
district_name: "Gampaha"
region: "Western Province"
population: 2300000
```

```
_id: ObjectId('680e91aa6a4e7558d7628378')
district_id: "D01"
district_name: "Colombo"
region: "Western Province"
population: 3000000
```

Shell Query:

```
db.districts.find().sort({ population:1 })
```

```
> db.districts.find().sort({ population:1 })
< {
  _id: ObjectId('680e922e6a4e7558d762837a'),
  district_id: 'D1003',
  district_name: 'Kandy',
  region: 'Central Province',
  population: 1500000
}
{
  _id: ObjectId('680e922e6a4e7558d7628379'),
  district_id: 'D02',
  district_name: 'Gampaha',
  region: 'Western Province',
  population: 2300000
}
{
  _id: ObjectId('680e91aa6a4e7558d7628378'),
  district_id: 'D01',
  district_name: 'Colombo',
  region: 'Western Province',
  population: 3000000
}
electionDB> |
```